PERIODIC RAW ATOMIC GAUSSIAN LIMITS ACROSS DIMENSIONS: DIMENSION-UNIFORM ANALYSIS AND LEAN INSTANCES AT $d = 2$–5 AND $d = 122409$

YJ ZHENG

Abstract. We study the fixed-volume continuum limit of the actual atomic lift of a massive periodic lattice Gaussian field. A Fourier argument proves, for every integer dimension $d \geq 1$ and every $s > d/2$, weak convergence in a weighted periodic Fourier realization of H^{-s}. We then separate this dimension-uniform analytic theorem from the dimension-specific Lean endpoints available in the public release: dimensions two through five, and the exceptional fixed dimension $d = 122409$. The latter is an unconditional whole-sequence theorem for the actual finite-source pushforwards in H^{-61206}, with compactness supplied by a stronger H^{-61205} energy estimate. Dimension-parameterized Lean lemmas establish weight summability and comparison facts, but the full convergence theorem is not parameterized by d. We record the source normalization, carrier conventions, target identification, proof dependencies, reproducibility commands, and explicit exclusions. In particular, no arbitrary-dimensional Lean convergence theorem, interacting limit, infinite-volume limit, or separate infinite-dimensional Gaussian predicate is claimed.

Contents

1. Introduction 1
1.1. Contributions and scope 2
1.2. Relation to the existing releases 2
2. Model and Fourier carrier 2
3. Dimension-uniform analytic theorem 3
3.1. Why the raw lift needs $s > d/2$ 4
4. Formalized dimensions and the large-dimensional instance 4
4.1. The 122409-dimensional theorem 5
4.2. Dimension-free Lean lemmas 5
5. Formalization, audit, and limitations 6
5.1. Pinned 122409 project 6
5.2. Reproduction and evidence classes 6
5.3. Limitations 7
Appendix A. Claim-to-evidence ledger 7
References 8
References 8

1. Introduction

Let $\mathbb{T}_L^d = (\mathbb{R}/L\mathbb{Z})^d$ be a periodic box with fixed side length $L > 0$. For mesh $a = L/N$, a centered Gaussian field on the periodic lattice $(\mathbb{Z}/N\mathbb{Z})^d$ is sampled and lifted without interpolation to the

Date: November 12, 2026.
2020 *Mathematics Subject Classification.* 60G15, 60B10, 68V15.
Key words and phrases. periodic Gaussian field, raw atomic lift, negative Sobolev space, weak convergence, Lean 4, formal verification.

atomic distribution

$$X_N = a^d \sum_{x \in (\mathbb{Z}/N\mathbb{Z})^d} \phi_N(x)\, \delta_{[ax]}, \qquad N = n+1.$$

The normalization is part of the statement: the site covariance is $a^{-d}(m^2 - \Delta_a)^{-1}$, where $m > 0$ and Δ_a is the nearest-neighbour periodic Laplacian. This is the raw atomic source, not a piecewise-constant interpolation or a spectral truncation.

The project now has two complementary kinds of result. First, a direct Fourier proof yields a theorem uniform in integer dimension. Second, Lean verifies concrete source-to-target convergence theorems at dimensions $2, 3, 4, 5$ and at 122409. These are not one dimension-quantified Lean endpoint. Keeping those claims distinct is central: the paper proof explains why the argument works for every d, while the theorem contracts and pinned projects state exactly which instances have been kernel-checked.

The unusually large dimension is useful as a stress test of the bookkeeping. At $d = 122409$, the target exponent $s = 61206$ lies strictly above $d/2 = 61204.5$, and the stronger source-energy exponent 61205 also lies above that threshold. The argument still uses the same shell-count and compact-tail mechanism; what changes is that every final convergence endpoint is instantiated at this single dimension.

1.1. **Contributions and scope.** The article gives: (i) the precise periodic model and weighted Fourier carrier; (ii) a dimension-uniform analytic proof for every integer $d \geq 1$ and $s > d/2$; (iii) a comparison table of the published Lean endpoints from two through five dimensions and the 122409-dimensional endpoint; and (iv) a reproducible evidence ledger binding the new high-dimensional project to its pinned Lean sources and query transcript.

The intended conclusion is weak convergence of probability measures on a separable weighted coefficient completion. It is not an interacting-field result. The finite-dimensional marginals of each limiting measure are centered Gaussian resolvent marginals; we do not promote this to a separately proved global ‘IsGaussian‘ predicate. Nor do we claim that all dimensions have been built as one generic Lean theorem.

1.2. **Relation to the existing releases.** The two-, three-, four-, and five-dimensional results have their own theorem declarations and previously published proof artifacts [2, 3, 4]. This paper is a synthesis and scope map, not a replacement for those sources. The new 122409-dimensional Lean project is included under `projects/D122409_RAW_GAUSSIAN`; it is pinned independently and imports only its declared Mathlib revision. The existing 5D endpoint remains distributed as a complete pinned source archive, linked from the repository README and listed in the ledger below.

Evidence boundary. The dimension-uniform theorem in Section 3 is proved analytically in this article. Its generality is not inferred from the finite list of Lean builds. Conversely, the five-dimensional and 122409-dimensional Lean results are source-specific, whole-sequence theorems; they are not merely numerical checks of the paper argument.

2. Model and Fourier carrier

For $N \geq 1$, write $a = L/N$ and identify lattice sites with $x \in (\mathbb{Z}/N\mathbb{Z})^d$. The positive operator is

$$(-\Delta_a f)(x) = a^{-2} \sum_{j=1}^{d} \big(2f(x) - f(x+e_j) - f(x-e_j)\big).$$

The source field is the centered real Gaussian vector with covariance

$$\mathrm{Cov}(\phi_N(x), \phi_N(y)) = a^{-d}(m^2 - \Delta_a)^{-1}(x, y).$$

The raw lift X_N is defined in the introduction. Its Fourier coefficient, with the physical torus convention, is

$$c_N(k) = a^d \sum_{x\in(\mathbb{Z}/N\mathbb{Z})^d} \phi_N(x)e^{-2\pi ik\cdot x/N}, \qquad k \in \mathbb{Z}^d.$$

For $r(k) = \|k\|_\infty = \max_j |k_j|$, set

$$w_s(k) = (1 + r(k))^{-2s}, \qquad \|c\|^2_{-s} = \sum_{k\in\mathbb{Z}^d} w_s(k)|c(k)|^2.$$

The carrier is the closed real subspace $c(-k) = \overline{c(k)}$ of the weighted complex ℓ^2 completion, represented isometrically by $J_s c = (\sqrt{w_s(k)}c(k))_{k\in\mathbb{Z}^d}$. At each fixed d this max-frequency norm is equivalent to the Euclidean Fourier Sobolev norm: $r(k)^2 \le \sum_j k_j^2 \le dr(k)^2$. The formula above, not the equivalence, is the precise carrier convention used in the Lean projects.

The discrete dispersion and its continuum counterpart are

$$\lambda_{N,d}(k) = m^2 + 4a^{-2}\sum_{j=1}^d \sin^2(\pi k_j/N), \qquad \lambda_d(k) = m^2 + (2\pi/L)^2\sum_{j=1}^d k_j^2.$$

Because $c_N(k)$ is periodic in each frequency coordinate modulo N, its covariance includes all aliases. The limiting real Gaussian Fourier family has

$$\mathbb{E}|c(k)|^2 = \frac{L^d}{\lambda_d(k)},$$

with the conjugacy constraint $c(-k) = \overline{c(k)}$. Equivalently, for real periodic test functions f, g and Fourier coefficients $\widehat{f}(k) = \int_{[0,L]^d} f(x)e^{-2\pi ik\cdot x/L}\,dx$,

$$\mathbb{E}[\langle X, f\rangle\langle X, g\rangle] = L^{-d}\sum_{k\in\mathbb{Z}^d} \frac{\overline{\widehat{f}(k)}\widehat{g}(k)}{\lambda_d(k)}.$$

Lemma 2.1 (Uniform mode estimate)**.** *For each $N \ge 1$ and $k \in \mathbb{Z}^d$,*

$$\mathbb{E}|c_N(k)|^2 \le L^d m^{-2}.$$

Proof. Diagonalize the periodic lattice operator by characters. The covariance eigenvalues are $a^{-d}\lambda_{N,d}(p)^{-1}$, where p is the residue of k modulo N. Orthogonality of the N^d characters and the raw-lift factor a^d give

$$\mathbb{E}|c_N(k)|^2 = \frac{L^d}{\lambda_{N,d}(p)} \le L^d m^{-2},$$

since every summand in the lattice dispersion is nonnegative. This calculation is finite-dimensional and exact; it retains the source normalization a^{-d}. □

Remark 2.2. Some machine-checked releases use a deliberately weaker upper bound with a factor 2 to simplify estimates. The present analytic proof uses the exact character covariance above. The Lean endpoint contracts and their own estimates, not this sharper paper bound, are the authority for claims about what those Lean declarations prove.

3. Dimension-uniform analytic theorem

Theorem 3.1 (Raw-lift convergence in every fixed dimension)**.** *Fix an integer $d \ge 1$, $L > 0$, $m > 0$, and $s > d/2$. Let μ_N be the law of the actual raw lift X_N in the carrier $H^{-s}_{\mathrm{Fourier}}(\mathbb{T}^d_L)$. Then there is a unique Borel probability measure $\gamma_{d,L,m}$ on that carrier with centered periodic-resolvent Gaussian finite Fourier marginals, and*

$$\mu_N \Longrightarrow \gamma_{d,L,m} \qquad (N = n + 1 \to \infty).$$

Proof. Choose r with $d/2 < r < s$ and write $\rho(k) = \|k\|_\infty$. The number of frequencies on the shell $\rho(k) = q$ is $(2q+1)^d - (2q-1)^d = O_d(q^{d-1})$, so $S_{d,r} = \sum_{k\in\mathbb{Z}^d}(1+\rho(k))^{-2r} < \infty$. By Lemma 2.1 and Tonelli,

$$\sup_N \mathbb{E}\|X_N\|_{-r}^2 \le L^d m^{-2} S_{d,r} < \infty.$$

Every finite atomic lift has bounded periodic Fourier coefficients, and hence lies in H^{-r} for $r > d/2$. The inclusion from the stronger coefficient norm H^{-r} into H^{-s} has compact closure on bounded sets: if P_M keeps only $r(k) \le M$, then

$$\|z - P_M z\|_{-s}^2 \le (1+M)^{-2(s-r)}\|z\|_{-r}^2.$$

Each P_M has finite-dimensional range. Thus stronger-energy balls have compact closure in the target carrier, and Markov's inequality gives tightness of (μ_N).

To specify the limit, choose one representative of each pair $\{k, -k\}$, and take independent standard real Gaussians ξ_0, ξ_k, η_k. Set

$$Z(0) = L^{d/2} m^{-1}\xi_0, \qquad Z(k) = \frac{L^{d/2}}{\sqrt{2\lambda_d(k)}}(\xi_k + i\eta_k), \qquad Z(-k) = \overline{Z(k)}.$$

Since $\mathbb{E}\|Z\|_{-s}^2 = L^d \sum_k w_s(k)/\lambda_d(k) < \infty$, the weighted sequence is almost surely in the carrier and defines a Borel random element. Its finite real Fourier families are centered Gaussian with the stated resolvent covariance.

For any fixed finite family of real sine and cosine evaluations, the lattice vector and its limit are centered Gaussian. Character orthogonality gives the exact joint covariance, and for each fixed mode k one has $\lambda_{N,d}(k) \to \lambda_d(k)$. For a fixed finite set of frequencies, distinct residue classes remain distinct modulo N once N is sufficiently large; the finite covariance matrices therefore converge entrywise to those of Z, and the corresponding finite-dimensional Gaussian laws converge. Tightness and Prokhorov's theorem yield a weakly convergent subsubsequence from every subsequence [1]. By the continuous mapping theorem, every subsequential limit has the same finite real Fourier marginals as Z. Finite coordinate cylinders generate the Borel sigma-algebra of the separable weighted ℓ^2 carrier, so these marginals determine the measure uniquely. All subsequential limits are therefore $\gamma_{d,L,m}$, which implies convergence of the full sequence. □

3.1. **Why the raw lift needs $s > d/2$.** The limiting field is smoother than the finite raw approximants: the extra resolvent factor yields finite expected H^{-s} norm already for $s > d/2 - 1$. The raw finite atomic lift, by contrast, has a frequency-periodic coefficient sequence. A nonzero periodic sequence does not decay at infinity, so it is generally not in H^{-s} at or below the summability threshold $s = d/2$. Thus the theorem's carrier condition is adapted to the actual approximants, not just to the limit law.

Evidence boundary. The preceding proof is a conventional mathematical proof within this article. Its finite-dimensional character computations and summability estimates are stated for variable d, but the cited Lean convergence theorems remain the finite list in Section 4. No source file is used to claim a universal Lean theorem.

4. Formalized dimensions and the large-dimensional instance

The Lean endpoints below are separate theorems. The Sobolev index in the table is the target index; the energy column records the stronger exponent used for compact containment, not a second convergence endpoint.

In every row, $L > 0$ and $m > 0$ are the only analytic hypotheses, $N = n+1$, and the source is the actual raw atomic Gaussian lift. The precise coefficient weights, source pushforwards, and target-identification conventions differ between the dimension-specific projects; the table is a summary, not a claim that all five use one common Lean carrier definition. The public 2D–4D root includes

TABLE 1. Whole-sequence Lean endpoints in the public release.

d	Target carrier	Strong energy	Source artifact	Status of claim
2	H^{-3}	H^{-2}	Native project modules	Lean endpoint
3	H^{-3}	H^{-2}	Native endpoint overlay	Lean endpoint
4	H^{-4}	H^{-3}	Native project modules	Lean endpoint
5	H^{-3}	$H^{-11/4}$	Pinned source archive	Lean endpoint
122409	H^{-61206}	H^{-61205}	Included standalone project	Lean endpoint

its endpoint theorems and audit contracts. The complete 5D pinned project is linked from the public release README as a source archive. The 122409 project is copied into this repository at `projects/D122409_RAW_GAUSSIAN`.

4.1. **The 122409-dimensional theorem.** Put $d = 122409$ and let the frequency index be $k : \mathrm{Fin}(122409) \to \mathbb{Z}$, equipped in Lean with the Pi supremum norm. The target weight and completion are

$$w_{61206}(k) = (1 + \|k\|)^{-122412}, \qquad \mathcal{X} = \ell^2(\mathrm{Fin}(122409) \to \mathbb{Z}; \mathbb{C}),$$

where the physical coefficient $c(k)$ is encoded as $\sqrt{w_{61206}(k)}c(k)$. The source-energy weight is $(1 + \|k\|)^{-122410}$. The strict inequalities

$$61206 > 122409/2, \qquad 61205 > 122409/2$$

place both exponents on the summable side. The Lean project proves summability by a dimension-parameterized product majorant and then specializes the actual-source arguments and final endpoint to exactly 122409.

Theorem 4.1 (Fixed 122409-dimensional raw-lift limit)**.** *For each $L > 0$ and $m > 0$, the actual fixed-volume raw atomic Gaussian probability measures converge weakly, as $N = n + 1 \to \infty$, on the weighted Fourier H^{-61206} completion to a unique Borel probability measure. Every finite real cosine/sine coordinate marginal of the limit is the centered periodic massive-resolvent Gaussian law.*

The exact declaration name and elaborated type are preserved in the included endpoint contract, which also queries the kernel axiom footprint. The principal theorem is defined in module `ActualWeakLimit122409`. Its proof chain includes measurability of the actual lift, the actual Gaussian pushforward, coordinate moments and the integral-series tail identity; compact closures of stronger-energy sublevels; uniform tightness and Prokhorov extraction; the normalized bridge from physical Fourier families to the target's weighted coordinates; convergence and identification of finite marginals; Borel generation by finite coordinate cylinders; uniqueness; and full-sequence convergence. The limit is selected as a probability measure on the actual target Borel space.

Remark 4.2 (Fractional stronger target)*.* The project also proves that a source weight with exponent 61204.75 is summable and that it has a positive decay gap to the 61205 source-energy weight. This supports a weight-level feasibility assessment for an H^{-61205} target, but does not prove convergence in that stronger target. The completed convergence theorem is in H^{-61206}.

4.2. **Dimension-free Lean lemmas.** The project also proves a dimension-parameterized weight-summability lemma for arbitrary natural dimension and real decay parameter $q > 1$, together with radial-to-product weight comparisons and one-dimensional axis-factor summability. These are genuinely dimension-generic helper results. They do not assemble a dimension-generic probability measure, compactness argument, finite-marginal limit, or whole-sequence theorem. The top-level convergence theorem is concretely instantiated at $d = 122409$; it is not obtained by applying a universal Lean convergence theorem.

Evidence boundary. The exact queried signatures and axiom footprints for the 122409 declarations are preserved in the included contract transcript. The pinned successful build and source/dependency manifest are in the same project. The 2D–5D theorem signatures and audit transcripts remain in their previously published project artifacts; see the release README and the ledger in Appendix A.

5. Formalization, audit, and limitations

The Lean and paper arguments have distinct validation roles. A successful Lean build establishes that the selected source files elaborate under their pinned toolchain and dependencies. A theorem contract queries exact declarations and their axiom footprints. A TeX build and PDF audit establish document production and artifact integrity. None of those checks, alone, substitutes for the other two forms of reasoning.

5.1. **Pinned 122409 project.** The standalone project in `projects/D122409_RAW_GAUSSIAN` pins Lean `v4.34.0-rc2`; the exact Mathlib revision is pinned in `lake-manifest.json`. Its successful build command targets `ActualWeakLimit122409`, and the endpoint contract is `EndpointContract122409.lean`. The supplied build log and endpoint transcript are preserved. The endpoint theorem and queried auxiliary declarations report only Lean/Mathlib foundational axioms `propext`, `Classical.choice`, and `Quot.sound`; there is no project axiom or `sorryAx` in the queried endpoint dependency contracts. This statement is limited to the checked declarations, not a theorem about every possible imported declaration in Mathlib.

5.2. **Reproduction and evidence classes.** From the repository root, run:

```
cd projects/D122409_RAW_GAUSSIAN
python audit/verify_manifest.py
lake build AImathPPhi2Clean.ActualWeakLimit122409
lake env lean EndpointContract122409.lean
```

The audit manifest binds every included Lean source, project configuration file and the retained build/contract logs by SHA-256. The contract prints endpoint signatures and calls `##print axioms`. To rebuild this article, run:

```
python paper/PERIODIC_GAUSSIAN_DIMENSIONAL_SWEEP/build.py
```

The script compiles with Tectonic and then independently inspects the resulting PDF.

Table 2. Evidence roles for the principal claims.

Claim	Primary evidence	Independent or artifact check
d-uniform analytic convergence	Proof in Section 3	Shell-count and covariance derivations are exposed; not a Lean theorem
2D–5D Lean endpoints	Existing public Lean declarations and pinned 5D archive	Their native typed contracts and build records; not rebuilt by this article's audit
$d = 122409$ actual-source convergence	Included Lean source and endpoint contract	Recorded successful pinned build, source manifest, and axiom transcript
Real-exponent weight threshold	Exact comparison $61205 > 122409/2$	Independent arithmetic check in the article audit
PDF delivery	Tectonic output	PyMuPDF text extraction, page-bound checks, title/scope assertions, embedded-font inspection

5.3. **Limitations.** All limits are at fixed positive physical volume and positive mass. The analytic theorem and formal endpoints do not establish an infinite-volume limit, an interacting field, reflection positivity, or Osterwalder–Schrader reconstruction. The 122409 Lean proof does not imply that the same Lean development works for arbitrary d. The dimension-parameterized summability lemmas address only weight comparison/summability inputs. The general weak-convergence statement is supported here by the written Fourier proof, with its analytic steps explicitly outside the Lean evidence claim.

The low-dimensional modules and 5D source bundle are preserved releases rather than dependencies of the included 122409 Lake project. Reproducing all of their proofs requires following the versioned commands in those artifacts. The article's audit checks its own text and PDF and the integrity of the included 122409 project; it does not re-execute the independent 2D–5D Lake builds.

Evidence boundary. The stated axiom footprint and successful build are reports about the included pinned evidence files. The PDF audit is not a proof checker. No numerical experiment is used to support the infinite-dimensional convergence theorem.

Appendix A. Claim-to-evidence ledger

Table 3. Evidence ledger for the article's theorem-level claims.

Claim	Evidence location	Classification / boundary
Uniform bound on one raw Fourier mode	Lemma 2.1; finite character diagonalization	Analytic derivation in this paper
Summability for $s > d/2$	Max-norm shell cardinality and convergent power series	Analytic proof in this paper
Tightness of actual laws	Stronger H^{-r} moment, compact inclusion and Markov	Analytic proof in this paper
Identification of the limit	Finite Gaussian covariance convergence and Borel cylinder uniqueness	Analytic proof in this paper
2D endpoint in H^{-3}	Public Lean modules and native contracts	Lean formalization; not rerun by this article
3D endpoint in H^{-3}	Published endpoint contract and 3D overlay	Lean formalization; finite real marginals identify the target
4D endpoint in H^{-4}	Public endpoint modules and release contract	Lean formalization
5D endpoint in H^{-3}	Pinned source archive and typed theorem contract	Lean formalization in standalone project
122409 endpoint in H^{-61206}	Included standalone project and queried endpoint contract	Lean formalization at this single fixed dimension
Arbitrary-d Lean weak convergence	No such endpoint is included or claimed	Explicitly not established by this article

For the high-dimensional row, `source_manifest.json` binds the included Lean sources and retained build/contract logs; the adjacent contract transcript records signatures and axiom footprints. The exponent audit is elementary: $2 \cdot 61206 = 122412$, $2 \cdot 61205 = 122410$, and both Sobolev orders exceed $122409/2$ where required. The optional fractional target assessment at 61204.75 does not change the theorem row.

Evidence boundary. The audit status attached to each row is deliberately not collapsed into a single "verified" label. In particular, the paper-level dimension-uniform proof and the fixed-dimensional Lean theorems are different evidence classes.

References

References

[1] P. Billingsley, *Convergence of Probability Measures*, 2nd ed., Wiley Series in Probability and Statistics, John Wiley & Sons, New York, 1999.

[2] Y. Zheng, *Periodic Gaussian lattice limits in two and four dimensions: a common proof mechanism and its Lean realization*, public source release, https://github.com/springoxn/pphi2-clean-finite-lattice/tree/main/paper/PERIODIC_GAUSSIAN_DIMENSION_BRIDGE.

[3] Y. Zheng, *From a dimension-uniform Fourier theorem to a formal three-dimensional periodic Gaussian limit*, with the H^{-3} endpoint supplement, public source release, https://github.com/springoxn/pphi2-clean-finite-lattice/tree/main/paper/PERIODIC_GAUSSIAN_3D_HMINUS3.

[4] Y. Zheng, *Five-dimensional actual-source periodic Gaussian limit in H^{-3}*, pinned Lean source archive and contract, public release tag https://github.com/springoxn/pphi2-clean-finite-lattice/releases/tag/v2026.11.11-5d-hminus3.